

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

Matteo Di Sante



EconophysicsLab offsite 2026

EconophysicsLab offsite 2026 | Slide 1

The statistics of soaring flight

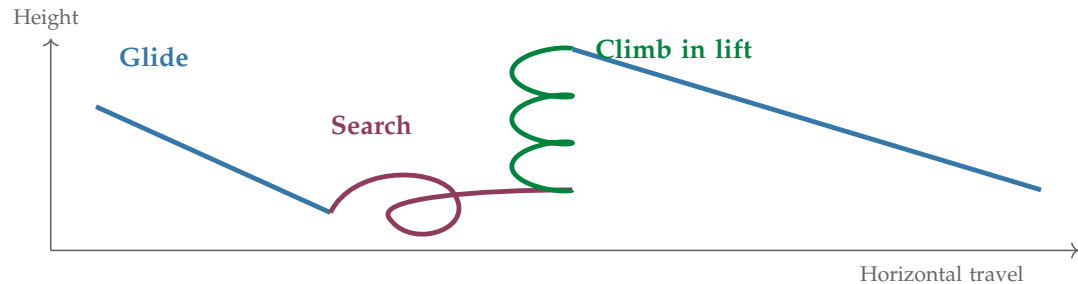
The aim is to describe soaring flight as a transport process and to identify empirical constraints for a later stochastic model. Today I cover the archive and its preparation, horizontal mean-square displacement and conditional comparisons through Section 3.2, followed by open questions and next steps.

The cover evokes the offsite setting: Lake Maggiore, Isola dei Pescatori and the Alps. A glide leads into local search and a helical climb in a thermal rising from a sun-warmed mainland slope. Upward filaments indicate airflow; the white helix is the aircraft path, with wind drift. The foreground island includes its bell tower and a red waterfront building without signs. This AI-generated illustration is geographically inspired, not a literal topographic map, measured airflow or a recorded flight.

Visual references: the user-supplied photograph and Comune di Verbania, Isola Superiore or Isola dei Pescatori, including the aerial photograph credited to Robin Ulrich.

Soaring is transport with an energy budget

Height is spent to travel and recovered in rising air.



Wind, terrain and route choice act throughout the flight.

Schematic

A flight is sustained by an energy budget: the pilot trades altitude for horizontal distance, searches for rising air, and recovers altitude in lift. These are useful behavioural descriptions, not phase labels supplied by the recorder or results of a segmentation shown here. Ground motion includes both air-relative motion and wind. The challenge is to explain whole-flight transport while respecting these physical constraints. The present measurements are the first layer of that programme. An MSD exponent alone cannot distinguish all the candidate mechanisms: persistent motion, changes of behaviour and heterogeneous conditions can produce similar second moments. The schematic is illustrative and its horizontal axis has no numerical scale. Source: thesis introduction, Sections 1.1–1.3.

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight

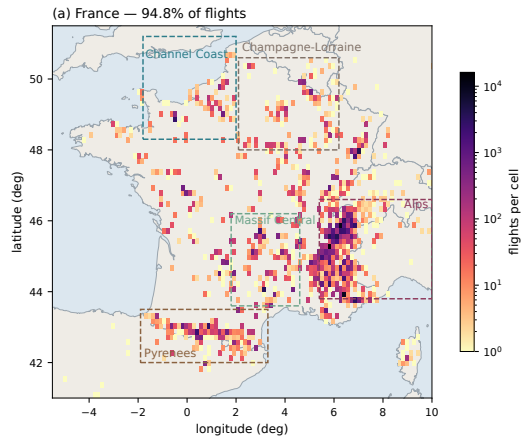
FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026



IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

The source is the FFVL Coupe Fédérale de Distance, with separate archives for paragliders and hang gliders. The verified downloads contain 186,052 and 6,716 tracklogs. Season names refer to their starting year. The IGC files supply time-stamped coordinates and altitude channels, while the catalogue supplies flight date, declared take-off site, route class and wing class. The map shows the retained population, not the raw downloads; both disciplines are pooled. About 94.8 percent of retained flights are in metropolitan France, with a strong Alpine concentration. Additional flights occur in La Réunion and elsewhere. This is a submitted cross-country archive with uneven geographic coverage, not a random sample of all free flight. Sources: thesis Chapter 2; generated/stats.tex, pipeline_census.tex and prelim_map.pdf.

Preprocessing: clean the recorded flight

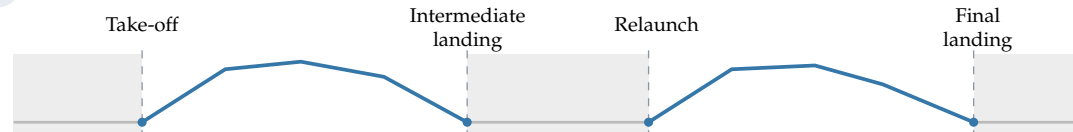
1 GNSS altitude

GNSS height $\longrightarrow$ Trajectory $z(t)$

Barometer $\longrightarrow$ Cleaning checks

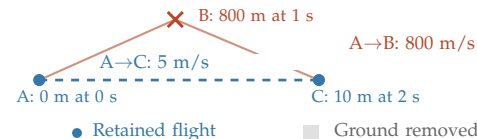
GNSS needs no pressure or temperature calibration.

3 Airborne interval



2 Fix-level cleaning

One fix: time, position and altitude.
Remove a faulty fix only when the remaining fixes can be physically rejoined.



Trim before take-off and after landing; excise witnessed stops and split the flight

GNSS supplies one altitude channel throughout, with an absolute-height reference that needs no pressure or temperature calibration. In supported paired records, high-frequency median spectra are comparable to barometry. This does not establish uniformly better absolute accuracy; logged vertical datums remain unaudited. Barometry corroborates cleaning decisions without replacing the output altitude.

A fix records time, position and altitude at one instant. In the example, A is at 0 m at 0 s, B at 800 m at 1 s, and C at 10 m at 2 s. Reaching B from A requires an impossible 800 m/s. Removing B leaves 10 m in 2 s: 5 m/s. Deletion therefore passes both the impossibility and rejoin tests. The dashed chord tests reachability; it is not interpolation. The pipeline handles contiguous blocks similarly. Without a timely reachable rejoin, it splits instead of deleting. A Hampel flag alone never deletes a fix; timestamp, frozen-lock and altitude defects have separate rules.

The timeline distinguishes take-off, intermediate landing, relaunch and final landing. Grey ground phases are removed; blue pieces remain segments of one parent flight. Outer boundaries use sustained speed. Interior stops also require eligible barometry supporting stable altitude: slow motion alone is insufficient. An excised stop becomes a boundary that later processing never bridges. Sources: thesis Sections 2.7–2.8 and cleaning ADRs.

4 Flight selection

Minimum flight thresholds

Blue: passes this check

Time ≥ 40 min

1.13% excluded at this check

Path ≥ 20 km

0.09% excluded at this check

Altitude range ≥ 600 m

2.68% excluded at this check

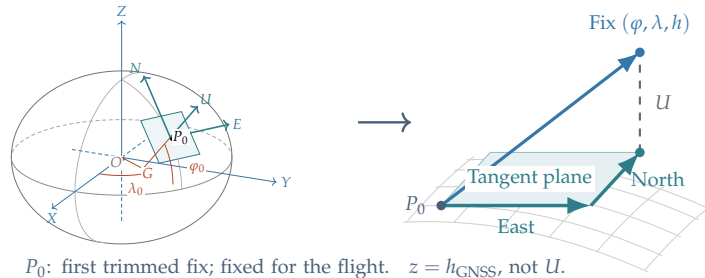
Plus integrity and reach checks

Retained after the full pipeline

156,406 paragliders 84.1%

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred position
 $\rightarrow$ local east-north distances in metres



P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

6,094 hang gliders 90.7%

Bars: pipeline census, pooled sequential checks. Coordinates: thesis §2.7, curvature exaggerated. Bottom: retained / downloaded, full pipeline.

Preprocessing: selection and local coordinates

The minima are 40 minutes, 20 kilometres and 600 metres of altitude range. Elapsed span is capped at 16 hours, with additional integrity and reach checks. Each bar pools both disciplines and represents the logs reaching that sequential check. Duration rejects 2,163 of 191,108 (1.13 percent), path rejects 174 of 188,942 (0.09 percent), and altitude range rejects 5,054 of 188,768 (2.68 percent). Three overlong logs leave before the path check. Blue marks the share passing that check. These are cascade rates, rather than independent tests on the raw archive.

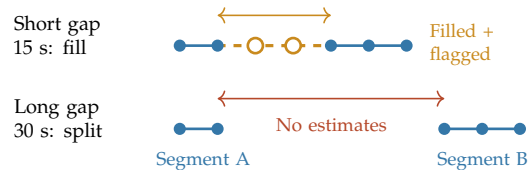
Latitude, longitude and logged GNSS height give an Earth-centred Earth-fixed position. Subtract the first trimmed fix, P_0 , then rotate into local east, north and up. The global schematic preserves the ellipsoid normal with exaggerated flattening. The close-up projects a later fix onto the fixed tangent plane. Keep the same origin and orientation for every segment of that flight. P_0 is near, but need not coincide with, true take-off.

The output vertical coordinate remains logged GNSS height, without subtracting origin height. Rotated Up contains curvature and is not used for vertical dynamics. Bottom counts cover the full pipeline, including resampling and smoothing: 156,406 of 186,052 paraglider logs and 6,094 of 6,716 hang-glider logs. Sources: thesis Sections 2.7–2.8, local ENU figure and pipeline_census.tex.

6 Time grid

Keep the flight's native cadence Δt .
Bridge only when $g \leq g_{\max} = \min(10\Delta t, 20 \text{ s})$.

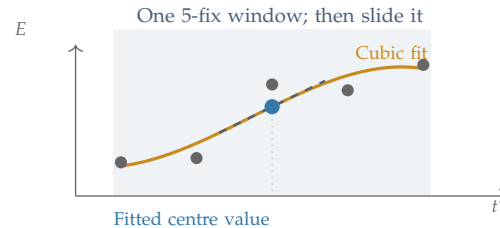
Example: $\Delta t = 5 \text{ s}$, $g_{\max} = 20 \text{ s}$



● Retained fix ○ Reconstructed fix

7 Savitzky–Golay fit

Fit a cubic locally to E , N , z .
Differentiate the polynomial analytically.



$$\widehat{x^{(n)}}(t_i) = P_i^{(n)}(0), \quad u = t - t_i$$

$n = 0$: position; $n = 1$: velocity; $n = 2$: acceleration

Analytic derivatives, without finite differences; fit each segment separately

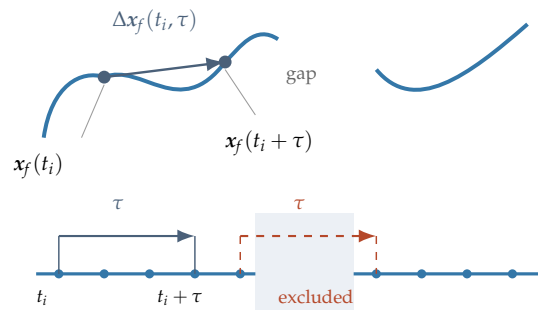
Preprocessing: time grid and smoothing

Stage 6 uses each flight's median recording interval. At a 5-second cadence the gap cap is 20 seconds. A 15-second gap has two internal grid fixes, linearly reconstructed and flagged: orange dashes and hollow markers. Blue markers are retained fixes in this aligned example. A 30-second gap leaves separate segments with no estimates between them. Mandatory cleaning boundaries always split. The same cap applies between finite altitudes: long altitude holes split the full trajectory. Reconstructed grid fixes are 0.58 percent for paragliders and 0.30 percent for hang gliders.

Stage 7 fits a cubic independently to E , N and z in a moving window. With physical time offset $u = t - t_i$, the fitted value and successive polynomial derivatives at zero give position, velocity and acceleration. Derivatives through order three are available from a cubic; the pipeline exports the first two. No subsequent finite differences are taken: those would amplify high-frequency position noise. The grey synthetic fixes and orange curve show an exact least-squares fit; the blue dot and dashed tangent mark its centre value and slope.

The 5-second configured timescale is converted to an odd window with at least five fixes. Five fixes span 4 seconds at 1-second cadence but 20 seconds at 5-second cadence. Smoothing can therefore affect short-lag statistics differently across cadences. Windows never cross a split; one-sided edge estimates and reconstructed support remain flagged. Sources: thesis uniform-sampling and smoothing sections; `resample.py` and `smoothing.py`.

One flight f , two retained segments



$\mathcal{I}_f(\tau)$: origin indices with both endpoints inside the same retained segment.

Pool origins within each flight;
then give each flight equal weight.

$$\Delta x_f(t_i, \tau) = x_f(t_i + \tau) - x_f(t_i)$$

$$m_f(\tau) = \frac{1}{|\mathcal{I}_f(\tau)|} \sum_{i \in \mathcal{I}_f(\tau)} |\Delta x_f(t_i, \tau)|^2$$

$$M_2(\tau) = \frac{1}{|\mathcal{C}(\tau)|} \sum_{f \in \mathcal{C}(\tau)} m_f(\tau)$$

$$\mathcal{C}(\tau) = \{f : |\mathcal{I}_f(\tau)| > 0\}: \text{contributing flights.}$$

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusion} \quad H = 1 \text{ ballistic motion}$$

Measuring horizontal transport

The index f labels a flight; i labels a candidate origin time t_i within it. The set $\mathcal{I}_f(\tau)$ contains only origins for which both endpoints lie in the same retained segment and the temporal-resolution rule supports the lag. We pool all such origins across the flight's selected segments. No displacement crosses a gap.

The upper drawing shows two pieces of one horizontal trajectory and a displacement within the first. The lower drawing shows their retained time intervals. Both arrows have the same lag, but only the arrow inside one segment contributes. The arrow crossing the grey gap is excluded even though its endpoints belong to retained segments.

The set $\mathcal{C}(\tau)$ contains flights with admissible origins. Its size gives the outer denominator; $|\mathcal{I}_f(\tau)|$ gives the inner denominator. Thus $n_f(\tau) = |\mathcal{I}_f(\tau)|$ and $N(\tau) = |\mathcal{C}(\tau)|$ if those symbols are used elsewhere. Every contributing flight has equal weight, regardless of its number of origins or segments. Displacement is a chord, not distance travelled.

The main analysis uses a common ten-second grid and fixed C10000 membership over 10–10,000 seconds. Slide 9 extends the short-lag view with cadence-limited admission. H is half the log–log slope: an effective growth exponent, not an identified self-similarity exponent. Source: thesis Section 3.1.1.

Why use a fixed population?

All available flights

As lag grows, short flights drop out.
MSD then mixes motion with a change
in which flights are averaged.

C_{10000} : select once for the largest lag

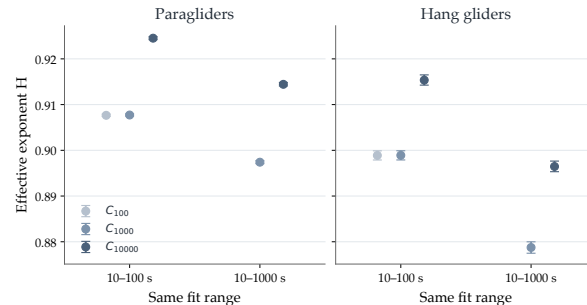
$10,000 \text{ s} \leq 0.8T_s$: $T_s \geq 12,500 \text{ s}$

Only these segments enter at every lag.

46,273 paragliders

2,326 hang gliders

Cost of selecting longer flights



H rises by about 2% on matched lag ranges.

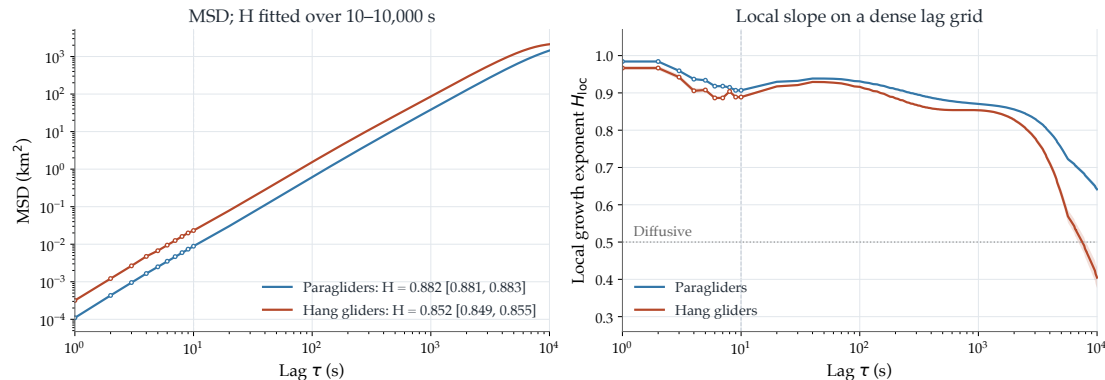
The available population loses short flights as lag increases. We therefore select a fixed set of flights and segments once, before computing the main curves. Requiring the maximum lag to be at most 80 percent of segment duration leaves at least one fifth of each selected segment available for origins at the longest lag. For ten thousand seconds, that means at least 12,500 seconds on the common grid and at least 251 origins at the maximum lag. Shorter segments of an otherwise eligible flight are excluded at every lag. The main cohort contains 46,273 paragliders and 2,326 hang gliders. The figure compares nested cohorts over identical fit intervals, using paired bootstrap draws. The main cohort increases H by 0.0164–0.0177, about 1.8–2.0 percent. We accept this small, measured duration-selection shift to keep the population constant across three decades of lag. Origins still vary with lag. Source: thesis Section 3.1.3 and ch3_transport_report.json.

Fixed flights and segments over 10–10,000 s; fewer origins at longer lags

Displacement growth from 1 s to 10,000 s

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C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.



Below 10 s the sample changes, so the local slope also reflects composition.

$$H_{PG} = 0.882 [0.881, 0.883]; H_{HG} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Displacement growth from 1 s to 10,000 s

The parent population is the original C_{10000} cohort, with the same selected long segments. Below ten seconds, a segment contributes when its native logging interval is no larger than the lag. Thus the contributing population grows from 33,450 paragliders and 690 hang gliders at one second to the full 46,273 and 2,326 at ten seconds. It stays fixed thereafter, through 10,000 seconds.

At short lags we use native origins and linearly interpolate endpoints when the lag is not a multiple of the logging interval, always within the selected segment span. This monotone admission rule differs from Figure 3.1: that figure uses the general population and requires exact native short-lag resolution, so its short-lag counts need not grow monotonically. From ten seconds, this slide uses the thesis's common ten-second grid and reproduces every original MSD point.

There are 277 evaluated lags, including every second from one to ten. Local H is half the log-log slope in a plus/minus 0.25-decade window, expanded to the nearest three lags when sparse. The one-second estimate uses one to three seconds. Slopes below ten seconds, and windows crossing that boundary, combine dynamics with changing sample composition. Lines connect computed estimates.

The original 10–10,000 second fits and 1,000 paired site-day bootstrap draws are preserved: $H=0.882$ [0.881,0.883] and 0.852 [0.849,0.855]. Bands are pointwise 90 percent intervals. Sources: thesis Section 3.1; cadence-support-report.json and measure_cadence_support.py.

Circuit type	Terrain / altitude	Experience
Declared route Open versus closed	Proxy: launch altitude Plains, hills, low and high mountains	Proxy: wing class Beginners: EN A/B/C Experts: EN D/CCC

Marginal comparison	Stratified comparison
Pool the other characteristics. Open vs closed, all altitudes together.	Keep a selected category fixed. Open vs closed, one altitude class at a time.

One factor at a time (OFAT): vary one input while holding the others fixed.
The archive comparisons leave other flight conditions free to differ.

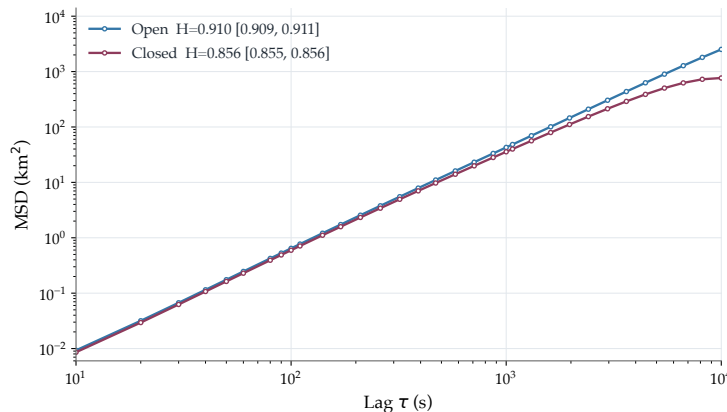
First compare groups; then test which contrasts persist across shared settings

Which flight characteristics shape MSD and H ?

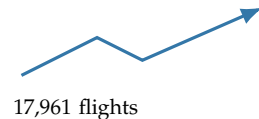
We examine three characteristics: terrain or launch environment, represented by the launch-altitude classes Plains, Hills, Low mountains and High mountains; pilot experience, with Beginners defined as EN A/B/C and Experts as EN D/CCC; and the declared circuit type, open versus closed. Launch altitude describes the starting environment rather than the terrain traversed throughout the flight. Wing class is an indirect indicator of experience and also reflects equipment performance.

OFAT has a precise experimental meaning: vary one factor while holding all other settings fixed. Presenting one recorded characteristic at a time is only informally similar. Our marginal comparisons pool the others, and our stratified comparisons hold a selected category fixed while remaining conditions can differ. They are useful for locating contrasts, checking whether they persist within settings, and detecting interactions. They do not isolate causal effects. The circuit-by-altitude and equipment-by-altitude comparisons include paired numerical contrasts, so their evidence goes beyond a visual reading. Source: NIST/SEMATECH e-Handbook of Statistical Methods, Section 5.2.1.2, <https://www.itl.nist.gov/div898/handbook/pri/section2/pri212.htm>, accessed 22 September 2026; thesis Section 3.2.

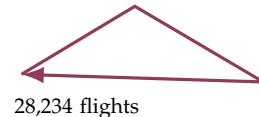
How do MSD and H differ by circuit type?



Open circuits



Closed circuits



$H_{\text{open}} - H_{\text{closed}} = 0.054$ [0.053, 0.056] (90% paired interval)
Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

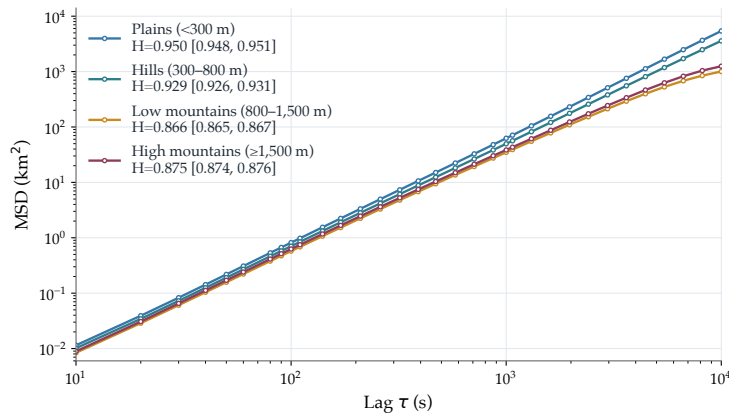
The catalogue describes the intended circuit, not a guarantee that the trimmed trajectory returns to its starting point. Closed routes include triangles and out-and-return flights. Pooling launch altitudes gives 0.910 [0.909, 0.911] for open routes and 0.856 [0.855, 0.856] for closed routes. Their paired difference is positive under the adopted bootstrap.

We fix the lag and average squared displacements over starting times along each segment. Suppose an ideal pilot reverses direction at minute 60, with the same speed on both legs. A two-minute interval from minute 59 to 61 spans the reversal and ends at its starting position; minute 20 to 22 covers straight flight. Thus even a short lag can capture a turn late in the flight. With equal leg times, $T/2$ and $T/3$ locate the first major turn along out-and-return and triangular routes. They do not define a minimum affected lag. Ten thousand seconds is the analysis limit; flight and leg durations vary. Testing the triangle hypothesis requires separating route types and examining actual leg durations.

The schematics illustrate geometry only. This marginal comparison pools launch environments and equipment, so its magnitude cannot rank the causal importance of route choice against terrain. Sources: thesis Section 3.2.1, ch3_conditional.json.

Why is pooled H higher in lowlands?

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Circuit composition

	Open	Closed
Plains	91.1%	8.9%
Hills	71.1%	28.9%
Low mtn.	33.5%	66.3%
High mtn.	35.0%	64.8%

Shares within each altitude group.
Unclassified: less than 0.3%.

Plains $>$ Hills $>$ Mountains in pooled H ; lowlands have larger open fractions.

The pooled altitude contrast also reflects circuit mix.

Initial GNSS altitude above sea level. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Why is pooled H higher in lowlands?

The pooled hierarchy is Plains 0.950 [0.948, 0.951], Hills 0.929 [0.926, 0.931], Low mountains 0.866 [0.865, 0.867] and High mountains 0.875 [0.874, 0.876]. Brackets give nominal 90 percent intervals. Both mountain classes lie below the lowland classes. Initial GNSS altitude is a broad launch-environment proxy.

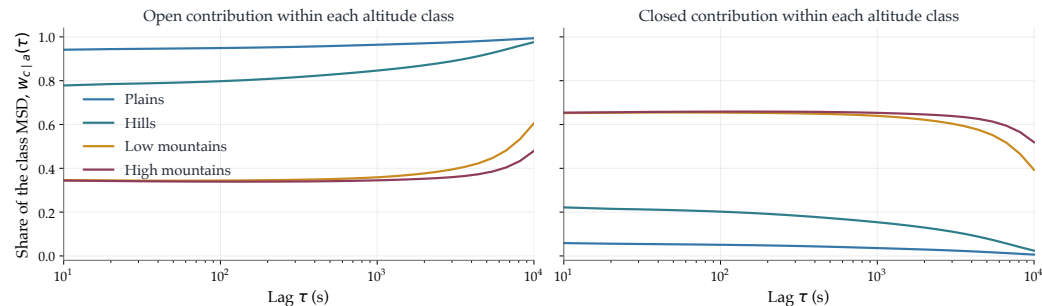
Read the composition table alongside the curves. Plains and Hills contain 91.1 and 71.1 percent open flights, compared with about one third in mountains. The preceding slide showed faster pooled growth for open routes. Thus this apparent altitude contrast also changes circuit composition; it cannot isolate terrain.

The next step is to show how flight fractions and circuit MSDs combine, then compare circuits within altitude classes. Do not infer which factor dominates from this marginal plot alone. The table includes 0, 1, 36 and 41 unclassified flights in its denominators. Source: thesis Section 3.2.2; ch3_conditional.json.

Circuit contributions change with lag

$$w_{c|a}(\tau) = \frac{p_{c|a} M_{2,c,a}(\tau)}{M_{2,a}(\tau)} \quad p_{c|a} = \frac{N_{c,a}}{N_a}$$

Within altitude class a ;
circuit c , flight fraction p .



Low mountains: 33.5% open flights supply 60.6% of the MSD at 10,000 s.

Changing MSD shares shape both the amplitude and fitted H of the pooled curve.

Observed shares; each class has its own denominator, including unclassified flights. No confidence bands.

For a fixed altitude class a , multiply each circuit's mean-square displacement by its flight fraction and divide by the full class MSD. This is the within-class contribution, not the contribution of altitude classes to all open flights. Open, closed and unclassified shares sum to one within each class and lag; the four coloured curves within one panel do not sum to one.

Open flights are numerous in Plains and Hills and have larger MSDs, producing 99.4 and 97.6 percent of their class MSD at 10000 s. In Low mountains, open flights are only 33.5 percent of flights but contribute 60.6 percent of the MSD at that lag. In High mountains, closed routes still supply about 51.9 percent. The unclassified contribution remains small and is explicitly retained.

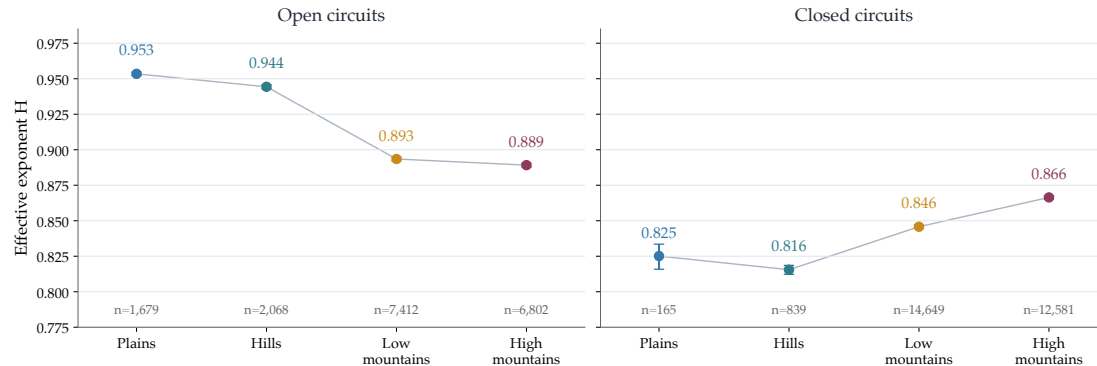
Flight fractions stay fixed, whereas MSD shares change with lag as circuit curves grow differently. These contributions shape the mixed MSD and its fitted H . A share is not an exact averaging weight for the H fitted over the whole interval. We retain no new local-exponent derivation. Source: thesis Section 3.2.2; circuit_msd_weights.py and ch3_conditional.json.

Does the circuit contrast persist across terrain?

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Does the circuit contrast persist across terrain?



Open minus closed (ΔH , 90% CIs): Plains 0.128 [0.120, 0.138]; Hills 0.129 [0.125, 0.132].
Low mountains 0.048 [0.046, 0.049]; High mountains 0.023 [0.021, 0.024].

Open: lowland H is higher. **Closed:** mountain H is higher.
Open > closed within every altitude class; the gap narrows in mountains.

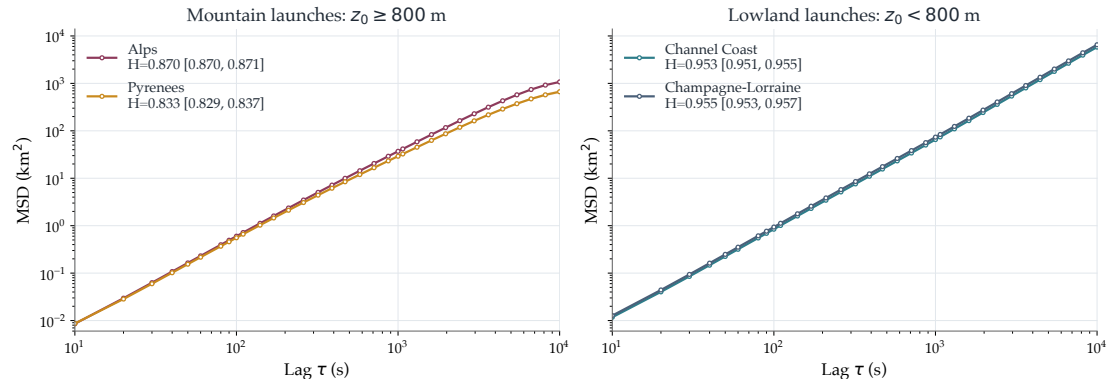
Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Read this figure in two directions. Comparing circuits within each altitude class gives open H greater than closed H in all four classes. The paired differences are 0.128 [0.120, 0.138] in Plains, 0.129 [0.125, 0.132] in Hills, 0.048 [0.046, 0.049] in Low mountains and 0.023 [0.021, 0.024] in High mountains. All four nominal 90 percent intervals are above zero. The circuit ordering survives altitude stratification, so different altitude composition alone cannot explain it. This is a consistent conditional association; its size still depends on altitude.

Comparing altitudes within each circuit reveals the reversal. Lowland-minus-mountain H contrasts are positive for open routes (0.051–0.064) and negative for closed routes (magnitudes 0.021–0.051). All eight paired intervals exclude zero. There is no general mountain penalty in H .

The two findings combine into a narrower circuit gap in mountains, as quantified by the four paired estimates and intervals on the slide. A physical interpretation is that terrain constrains how pilots realise either objective. Open routes may lose outward persistence; longer legs or later returns could make mountain closed routes less affected by closure within the fitted interval. These mechanisms need route and weather measurements. The conditional orderings and their interaction are already measured; future factorial analysis adds equipment and further interactions. Sources: thesis Figure 3.9 and ch3_transport_task_table.tex.

Is launch altitude enough to describe transport?



ΔH : Alps–Pyrenees 0.037 [0.033, 0.041]; lowlands -0.002 [-0.005 , 0.001].

Broad altitude classes do not capture all transport differences.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

This comparison holds the broad altitude setting fixed within geographic boxes. The Alpine and Pyrenean groups retain only launches at or above 800 metres; the two lowland groups retain only launches below 800 metres. There are 32,790 Alpine flights, 2,257 Pyrenean flights, 1,068 Channel Coast flights and 798 Champagne-Lorraine flights. Alpine H is 0.870 [0.870, 0.871] versus 0.833 [0.829, 0.837] in the Pyrenees; their paired difference is 0.037 [0.033, 0.041]. The lowland exponents are 0.953 [0.951, 0.955] and 0.955 [0.953, 0.957], with difference -0.002 [-0.005 , 0.001]. A similar exponent does not imply identical motion: displacement amplitudes and curve shapes can differ. These selections do not match weather, route composition, exact altitude, site or equipment. The regional contrast is descriptive and does not identify the effect of mountain geometry or thermal triggers. To examine the spatial mechanism directly, the next slide compares climb locations at two heights in selected mountain and plains cells. Source: thesis Section 3.2.3; ch3_conditional.json.



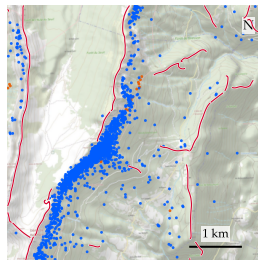
All dates and years pooled. Vilpellet climb crossings.

H2: Chartreuse, Alps

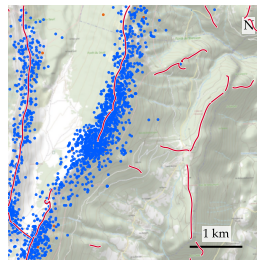
High mountains #2 / 15,840 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,856 crossings / 1,261 flights



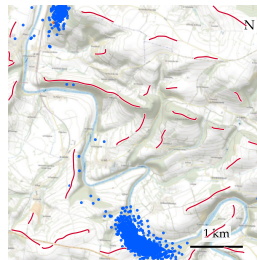
1,868 crossings / 1,452 flights

P1: Suisse Normande, Channel Coast

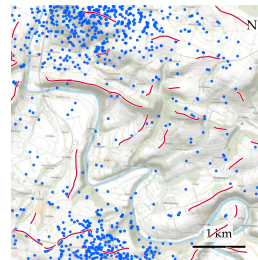
Plains #1 / 2,246 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,441 crossings / 734 flights



901 crossings / 574 flights

100 to 600 m: P1 climbs spread out; H2 climbs stay near slopes and ridges.

Observed lift use is less confined aloft in P1, supporting more spatial freedom for route choices.

Blue/orange: paragliders/hang gliders. Red: approximate IGN DEM-derived crests. AGL: fixed cell reference. © IGN Plan IGN / RGE ALTI.

How do climb locations change with height?

The two five-kilometre viewer cells are H2, High mountains number 2 near Aulp du Seuil in the Chartreuse, Alps; and P1, Plains number 1 in inland Suisse Normande within the Channel Coast box. P1 contains local slopes despite its low starting-altitude class. Both disciplines and all available dates through June 2026 contribute, without seasonal or MSD-duration cuts.

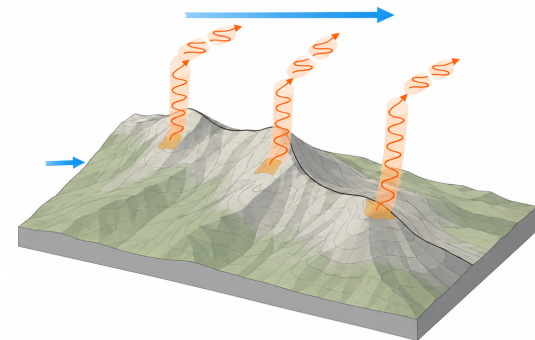
Heights are offsets from fixed references: 1756 m ASL in H2, based on five screened starts, and 185 m in P1, based on 1165. They are not local terrain clearance. Red crests are independently derived from IGN RGE ALTI at 25 m sampling with 50 m smoothing. These scale-dependent traces include secondary ridges and retain gaps at weak or ambiguous junctions. They do not move with the flight plane.

H2's two main bands at 100 m chiefly follow outer slopes of separate ridges: Lances de Malissard and Aulp du Seuil-Bellefont, enclosing the Vallon de Marcieu. They do not simply flank one central crest. In P1, crossings cluster near local slopes at 100 m and occupy a broader area at 600 m; H2 retains narrower terrain-associated corridors.

This is observed lift use, not tracked thermal centres. Possible slope lift and different flights and weather contribute at each height. Source distribution and wind advection are proposed explanations on the next slide; neither wind profiles nor their contribution to H are measured here. Sources: prepared viewer intersections: IGN Plan IGN and RGE ALTI. Licence Ouverte 2.0.

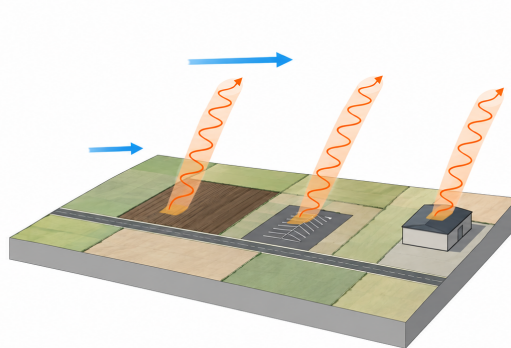
Mountains: sources on heated slopes

Possible shelter below crests; stronger shear aloft.



Lowlands: separated heated surfaces

Dry field, asphalt, warm roof; drift from the base.



Distributed sources and variable wind could broaden lowland crossings.

Mountain slope sources and possible shelter can keep lift use closer to terrain corridors.

Orange: rising air; blue: wind. Illustrative profiles, not measurements. Sources: FAA, Ch. 9–10; SSA, Lift Sources.

Why do lowland climb locations spread aloft?

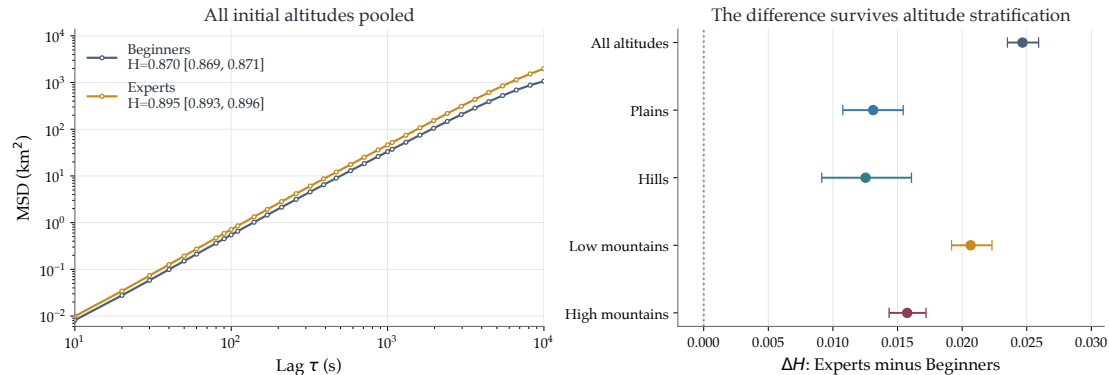
Distributed heat sources and wind advection could explain the stronger broadening between 100 and 600 m in P1. Slope-associated sources and possible shelter could preserve a narrower mountain footprint. Mountain sources can form on heated slopes and move upslope before detaching; they need not originate on crests. Dry fields, asphalt and roofs illustrate separate potential lowland sources, not guaranteed thermals. Orange is buoyant rising air, blue is horizontal wind. Helices indicate ascent, not measured rotation; fragments illustrate possible shear disruption.

Sheltered slopes with stronger flow above the crest are one possible mountain profile. Exposed lowland sources may drift from lower heights. Neither profile is measured or universal: valleys can channel strong wind, and inland P1 is not necessarily coastal.

For a steady idealised plume, $\Delta r_h \simeq (u_h/w)\Delta z$, where w is upward air speed, not glider climb rate. Shared wind and rise speed translate a source pattern; variability across dates can broaden pooled crossings. Matched climbs and wind profiles are needed to test this.

Terrain guidance and clearance could interrupt open routes' outward progress. Longer legs or later returns could weaken closure's signature in closed routes. Both may narrow the circuit gap; route timing remains untested. Sources: FAA Glider Flying Handbook, Chapters 9–10; SSA, Lift Sources; Schroeder and Buck, Fire Weather, Chapter 6; thesis Section 3.2.2.

Experts and beginners across altitudes



Higher H for experts in every class; the gap depends on altitude.

A constant speed advantage changes MSD amplitude, not H .

Wing class is a proxy. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

The Beginners group contains 27,609 flights and the Experts group 16,497; 2,167 flights are excluded from this comparison. These are the operational groups defined on slide 10. Individual experience is not observed: an experienced pilot can fly a lower-class wing, and the same pilot can contribute to different groups on different flights. The pooled exponent difference is 0.025 [0.024, 0.026]. Within altitude classes the differences are 0.013 [0.011, 0.015], 0.013 [0.009, 0.016], 0.021 [0.019, 0.022] and 0.016 [0.014, 0.017], all with paired intervals excluding zero. The Low mountains contrast exceeds the others in the paired interaction tests. A constant speed multiplier alone changes MSD amplitude, not its log-log slope, so the difference concerns the lag dependence too. The altitude-dependent gap is an equipment-by-altitude interaction already tested with paired contrasts. Each cell still pools circuits and weather. Circuit and equipment contrasts both depend on altitude, motivating the joint model with interaction terms discussed in the future-work section and the appendix. Source: thesis Section 3.2.4, ch3_conditional.json and ch3_conditional_equipment_table.tex.

Segmentation

Vilpellet's HMM: developed on tracks at ≥ 1 Hz.
Slower logs and some missed climbs motivate a revision.
Next: continuous observations and timing in physical seconds.

Anisotropy

Mountains: test alignment with the mountain-chain axis.
Plains: test weaker alignment with mean wind.
Preliminary observations; quantify under matched conditions.

Distribution scaling

Collapse tests and moment scaling currently give mixed indications.
A power-law MSD alone does not establish self-similarity.

Validate across logger cadences before interpreting phase statistics and scaling

Work in progress: resolving phases and directions

The current Vilpellet segmentation is useful and already supplies the thermal-plane examples. The source study retained trajectories at a minimum frequency of one hertz; its appendix uses binary trajectory features. That domain does not validate transfer to our slower loggers. Visual inspection also suggests some climbing or lifting stretches are missed. These are diagnostics, not a completed accuracy benchmark. The proposed replacement remains an HMM with discrete glide, search and climb states, but continuous-valued observations such as vertical speed and turning measures. Features must be defined in physical time and state persistence must account for sampling interval. Merely replacing binary emissions with Gaussians does not solve cadence dependence. Validate on reviewed trajectories and deliberately downsampled high-frequency tracks.

For anisotropy, estimate the displacement tensor at each lag and compare its main axis with local ridge orientation or measured wind. Mountain alignment and weaker lowland wind alignment remain preliminary. Terrain and wind directions must be measured independently of the trajectories.

Distribution-collapse and moment-scaling diagnostics are not yet consistent enough to claim or reject a common self-similar law. Repeat them on matched populations, lag ranges and directions with uncertainty. The fitted MSD exponent is only a second-moment summary.

Next: gliding and waiting

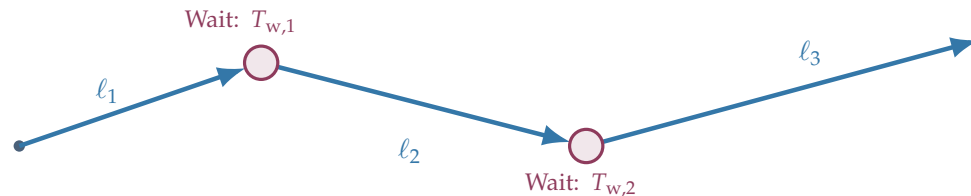
A two-phase model, motivated by Vilpellet et al.

Displacement: gliding

Duration T_g , length ℓ and direction θ .
Retain turning-angle memory between glides.

Waiting: search + thermalling

T_w : total time between successive glides.
Estimate one pooled waiting-time distribution.



Waiting neglects local search motion and thermal drift: a modelling approximation to test.

Monte Carlo: test MSD, displacement distributions, anisotropy and memory

The proposed transport model has two phases: displacement during gliding, and waiting that combines search and thermalling. Vilpellet et al. report ballistic glides, subdiffusive search and slow horizontal drift during climbs. These observations motivate our reduction; they do not establish a fitted two-state model.

Retain three behavioural labels for segmentation, then merge each contiguous search/thermalling block into one waiting episode. Its duration includes all such intervals until the next glide, allowing repeated alternations or an absent search phase. Estimate total waiting times directly, without assuming independent search and climb durations.

Glides generate displacement over finite durations. Estimate their lengths, durations, directions and successive dependence, including turning-angle memory. In the simplest model, position stays fixed during waiting while time advances. This neglects local search motion and thermal advection: quantify the displacement error over the target scales. Reorientation between glides remains part of the model.

Simulate in two dimensions plus time, using the observed sampling and estimators. Test MSD, displacement distributions, anisotropy and temporal correlations on held-out flights. Matching one H is insufficient. Source: Vilpellet et al. (2026), Sections IV and VI, Figures 3–4.

Next: social flight and the spatial supply of lift

Solo versus grouped flights

Build reliable group / solo labels.
Compare transport within shared terrain,
circuit and wing-class settings.

$$\Delta H_{a,c,e}^{\text{social}} = H_{a,c,e,\text{group}} - H_{a,c,e,\text{solo}}$$

Factorial models: estimate this contrast
and how it changes across the other factors.

Where is usable lift?

Estimate the horizontal distribution of ther-
mals
conditional on height and weather.
Account for where gliders actually sampled.



Test routes against an explicit objective:
for example, maximise expected progress
subject to altitude and safety constraints.

Reliable detection of simultaneous group flight is a prerequisite. For each altitude, circuit and equipment setting, compare grouped with solo flights, then model how that contrast varies between settings. Adding a binary social factor doubles the four-by-two-by-two table to 32 cells. Factorial models include both conditional contrasts and interactions with terrain, circuit or equipment; adequate cell support and dependence-aware resampling remain essential. The current conditional comparisons already demonstrate circuit-by-altitude and equipment-by-altitude associations. The proposed joint model brings these factors together; it does not turn an observational archive into a randomised experiment or guarantee causal identification. The symbolic model is retained in the appendix.

The second direction estimates the spatial availability of usable lift from climb data, conditional on altitude and weather. Crossing density is biased by routes, exposure and pilot choices, so it is not directly thermal density. Correcting for flight sampling and, where possible, using atmospheric information is necessary. Use the resulting field in a routing model only after stating what optimal means: expected progress, travel time or another criterion, subject to altitude, reachability and safety constraints. This is a future scientific model, not flight-navigation advice or a validated operational route planner.

Separate social contrasts from composition; link route choices to the available lift

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.
parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

IGN / Géoplateforme: Plan IGN map backgrounds.
IGN RGE ALTI: local terrain windows and derived crests.
Licence Ouverte 2.0; only the selected cells downloaded.

Locator and analysis

Natural Earth: France outline, public domain.
MSD, intervals and crossings: this thesis's prepared pipeline.
Code, figure scripts and recorded source provenance

Data and geographic sources

The FFVL archive supplies the observed trajectories and catalogue metadata. Paraglider and hang-glider collections have separate portals and season ranges, shown earlier. The pipeline applies the documented cleaning, selection and equal-flight estimators; generated figures are outputs of this thesis rather than independent external datasets. Thermal crossings come from the prepared viewer store.

IGN Plan IGN is the coloured map background. Approximate crest lines are computed independently of flight positions from RGE ALTI terrain, requested at 25 m sampling for twelve selected five-kilometre cells with a 500 m processing buffer. The total terrain download is about 2.64 MiB. The source licence and precise queries are retained with the extracts. The France locator uses Natural Earth country boundaries in the public domain.

Named summit points from IGN BD TOPO were tested earlier but are not the crest overlay currently displayed. The thermal-mechanism image is a schematic, not an atmospheric measurement. Source hashes, extraction parameters and figure scripts are recorded with the presentation and thesis.

Vilpellet, Darmon & Benzaquen (2026). *From Random Walks to Thermal Rides: Universal Anomalous Transport in Soaring Flights*. arXiv:2601.01293.
Transport, HMM phase segmentation and phase-duration benchmarks.

NIST/SEMATECH. *e-Handbook of Statistical Methods*,
§5.2.1.2: One Variable at a Time; §5.3.3.3: Full Factorial Designs.
Mohr, Wilson & Freund (2022). *Statistical Methods*, 4th ed., §9.3, §9.7.

FAA (2024). *Glider Flying Handbook*, Ch. 9–10.
Soaring Society of America. *Lift Sources*.
Schroeder & Buck (1970). *Fire Weather*, Ch. 6: General Winds.

Savitzky & Golay (1964). Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Analytical Chemistry* 36, 1627–1639. doi:10.1021/ac60214a047.

Selected bibliography

Vilpellet, Darmon and Benzaquen provide the central transport and segmentation comparison. Their reported phase durations are a benchmark to test after our own segmentation is validated. NIST and the statistical-methods textbook support the distinction between controlled one-factor-at-a-time experiments and factorial models with interactions. The FAA, SSA and general-wind sources support possible lift and wind mechanisms, not a measured wind profile for either selected cell. Savitzky and Golay give the local-polynomial basis of the smoothing and derivative estimates. The main talk keeps the statistical results, physical interpretations and proposed mechanisms distinct. The full references, publication details and links are maintained in thesis/references.bib.

What the data show

Persistent displacement growth.

A fixed population makes lag comparisons interpretable; H summarises one fit range.

Circuit and terrain interact.

Open > closed within every altitude class. The altitude ordering reverses by circuit.

Composition matters.

Group MSD shares change with lag; flight counts alone cannot explain pooled H .

What remains open

Mechanisms and uncertainty.

Terrain-linked lift is observed; wind and route mechanisms need matched tests.

Cadence and scaling.

Validate segmentation, anisotropy, distribution scaling and bootstrap coverage.

Joint models and social flight.

Quantify conditional group / solo contrasts; test a phase-based stochastic model.

Take-home messages and open questions

The main contribution is a reproducible statistical description, followed by marginal and stratified comparisons that reveal which contrasts survive changes of setting. The fixed cohort removes lag-dependent entry and exit of flights over the principal fit range, while preserving the need to discuss duration selection and statistical uncertainty.

Open trajectories have larger effective exponents than closed trajectories within every launch-altitude class. The altitude ranking reverses between open and closed circuits: mountain flights do not have a universally smaller H . Unequal circuit composition and lag-dependent MSD contributions explain why a pooled terrain plot alone is insufficient. Wing-class contrasts also depend on altitude, although wing class mixes equipment and an experience proxy.

Thermal-plane maps show terrain-associated climb corridors in H2 and stronger broadening with height in P1. They support a physical interpretation but do not isolate wind or explain the circuit interaction by themselves. Next steps validate segmentation and uncertainty, resolve anisotropy and distribution scaling, and combine drivers in joint factorial models, especially for solo versus grouped flight. Phase durations and dependence will then constrain and test a stochastic model. These are open tasks, not further established results.

Next: test the mechanisms with joint models and validated flight phases

Thank you

Questions and discussion

Thanks to the pilots sharing their flights,
the FFVL archive and the research community.

Matteo Di Sante

EconophysicsLab offsite 2026

Thank you

Thank the audience and open the discussion. Acknowledge the pilots who contributed their flight logs, the FFVL archive making those records available, and the research community whose methods and results support the analysis. The quantitative factorial example and the bootstrap-method slide follow as backup material.

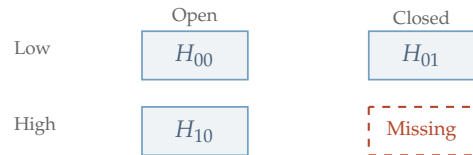
A: 4 altitudes C: 2 circuits E: beginners / experts $4 \times 2 \times 2 = 16$ cells

$$H_{ace} = \underbrace{\mu + A_a + C_c + E_e}_{\text{additive part}}$$

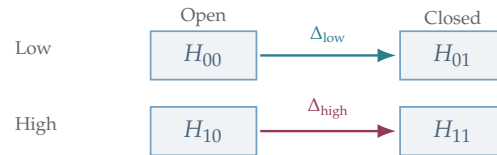
$$+ (AC)_{ac} + (AE)_{ae} + (CE)_{ce} + (ACE)_{ace}$$

Example: two altitude levels and two circuits; beginners / experts group held fixed.

OFAT: one combination missing



Full factorial: all four combinations



Interaction: $D_{AC} = \Delta_{\text{high}} - \Delta_{\text{low}} \neq 0$ means the circuit effect depends on altitude

Free coefficients:
6 additive
10 interactions

Sum-to-zero constraints

Appendix: factorial models and interactions

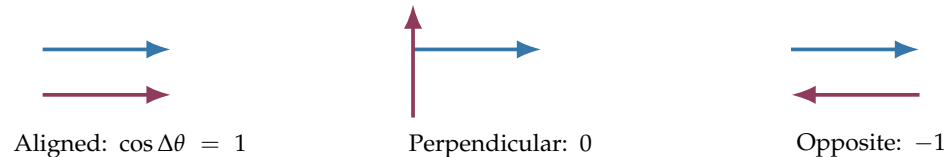
Fit H to each cell MSD over the same lag interval. Under sum-to-zero constraints, the additive model has six free coefficients: one mean, three altitude contrasts, one circuit contrast and one experience-group contrast. The ten interaction coefficients are three AC, three AE, one CE and three ACE. This saturates sixteen cells and supplies no residual error estimate by itself; weighting and uncertainty remain under review.

The diagrams fix experience group and code altitude and circuit as $x, z = 0, 1$. Rows are low/high altitude; columns are open/closed circuit. OFAT at (0,0) observes (0,0), (1,0), (0,1) and misses the high/closed cell. In $H_{xz} = \beta_0 + \beta_A x + \beta_C z + \beta_{AC} xz$, the interaction term is zero in those three settings, so its coefficient is undetermined.

The full grid supplies the fourth cell. The arrows give $\Delta_{\text{low}} = H_{01} - H_{00}$ and $\Delta_{\text{high}} = H_{11} - H_{10}$. Their difference is $D_{AC} = \beta_{AC}$. Additivity implies zero: the circuit effect is unchanged by altitude. A nonzero estimate requires uncertainty assessment. These cells are symbolic, not new measurements.

Archive comparisons are observational, not controlled OFAT. Binary group/solo doubles the table to 32 cells and adds its interactions. Reliable social labels, cell support and resampling for shared conditions are needed. No provisional factorial results are shown. Sources: NIST/SEMATECH, Sections 5.2.1.2 and 5.3.3.3; Mohr, Wilson and Freund (2022), Sections 9.3 and 9.7; thesis Section 3.2.5.

$$\mathbf{v}(t) = s(t) \mathbf{e}(t), \quad \mathbf{e} = (\cos \theta, \sin \theta) \quad \Delta\theta = \theta(t + \tau) - \theta(t)$$



$$C_v(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle s^2 \rangle} = \frac{\langle s(t)s(t + \tau) \cos \Delta\theta \rangle}{\langle s^2 \rangle}$$

$$C_\theta(\tau) = \langle \mathbf{e}(t) \cdot \mathbf{e}(t + \tau) \rangle = \langle \cos \Delta\theta \rangle \quad \text{Constant speed: } C_v = C_\theta.$$

Persistent mean drift can leave a nonzero baseline; also inspect centred velocity covariance.

C_v combines speed and direction; C_θ isolates directional agreement

Appendix: velocity and directional memory

Write horizontal velocity as speed times a unit heading. The dot product is the product of the two speeds multiplied by the cosine of their angular difference. Averaging and dividing by mean squared speed gives the displayed normalised velocity autocorrelation under stationary sampling. Aligned directions contribute positively, perpendicular directions give zero and reversals contribute negatively. Speed products weight those contributions, so velocity autocorrelation contains directional information but does not isolate it.

Correlating unit vectors removes speed weights: $C_\theta = \langle \cos \Delta\theta \rangle$. At constant speed, $C_v = C_\theta$. With varying speed they generally differ; speed and angular averages cannot be factorised without additional assumptions. Headings require nonzero speed.

Mean drift, including wind, can leave a positive long-lag baseline without memory in the fluctuations. The centred autocovariance is $K_v(\tau) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle - |\langle \mathbf{v} \rangle|^2$. It describes fluctuations around drift, whereas unit-vector correlation describes the original headings. State which convention is used.

For stationary velocities, writing $R_v(\tau) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle$ gives $M_2(\tau) = 2 \int_0^\tau (\tau - u) R_v(u) du$. This connects velocity dependence to displacement growth, but a fitted H does not specify that dependence uniquely. Use physical-time lags, within-segment pairs and consistent weighting. This slide derives identities; it does not report measured correlations.

Appendix: bootstrap confidence intervals

1 Group flights

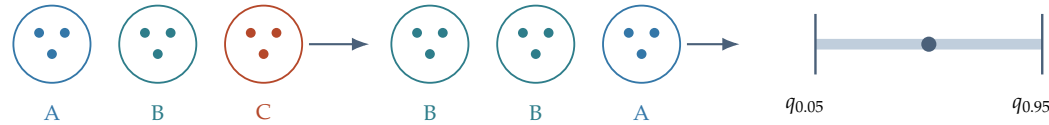
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

The reported confidence intervals use the current site–day cluster bootstrap. Each group combines flights from the same declared take-off site and calendar day. We sample groups with replacement, preserve all flights belonging to a sampled group, recompute the equal-flight estimator, and repeat this one thousand times. The same multiplicities are paired across lags and conditional comparisons. The 5th and 95th percentiles form nominal 90 percent intervals; for a curve they are pointwise, not a simultaneous confidence band.

Clustering aims to keep dependent flights together and make different groups approximately independent. The current partition is too fine: nearby sites and neighbouring days can share weather and other conditions, so flights in different groups can still be dependent. In the current cohort, 71.1 percent of paraglider groups and 81.3 percent of hang-glider groups are singletons. The bands are consequently likely to understate uncertainty; their nominal level should not be read as demonstrated coverage.

Validation with 50-km cells crossed with initial-altitude class and month of year gives standard errors for fitted H roughly four times the site–day value for paragliders and twice for hang gliders. Further aggregation changes the errors again; independence and uncertainty have not stabilised. These are diagnostic comparisons, not a validated replacement partition, and other chapter bands have not been recomputed with that scheme. All intervals in this presentation retain the original bootstrap convention. Sources: thesis Section 3.1.2; ch3_transport_grid_bootstrap.json and its generated values.